Geometry Analysis of MNIST Digits Using PCA, UMAP, and CCA

Alisha Pham

Tianye Dou

1 Introduction

This project investigates structural properties of the MNIST dataset using Principal Component Analysis (PCA), Uniform Manifold Approximation and Projection (UMAP), and Canonical Correlation Analysis (CCA). Our goal is to understand what these methods reveal about within digit variation, global manifold structure and symmetry across digit classes.

Our first question examines how the leading PCA directions differ across digit classes, chosen to identify the primary geometric features that drive within digit variation. The second question compares how PCA and UMAP represent the global MNIST manifold in three dimensions and how UMAP parameters affect digit separation. We study this because PCA and UMAP rely on different assumptions, global linear variance versus local neighborhood structure, and comparing their embeddings clarifies how these assumptions affect digit separation. The third question uses top–bottom CCA to assess how strongly the upper and lower halves of each digit image correspond, since CCA provides a quantitative way of measuring symmetry, a natural property of handwritten digits.

2 Data Summary

The MNIST dataset used in this project contains 5000 images of handwritten digits (0–9), with 500 images per class. Each 28×28 image is represented as a 784-dimensional vector with pixel intensities in [0,255]. We preprocess the data by standardizing both the full dataset and each digit subset.

To summarize the dataset, we compute a mean pixel heatmap and report the mean and standard deviation of zero valued pixels for each class. The heatmap captures the average structure of each digit, while the zero pixel mean quantifies the proportion of background relative to the digit. The standard deviation reflects how consistently each digit occupies space across samples.

2.1 Mean Pixel Heatmap

From Figure 1, several digits (0, 1, 4, 5, 8, 9) show varying degrees of rightward tilt. Digit 1 exhibits faint horizontal marks at the top and bottom, showing that it is not always written strictly vertically. Digits 4, 7, and 9 display a subtle bottom bar, reflecting variation in where writers position the digit and how they complete the final stroke. Digit 7 also shows a faint initial hook, indicating additional stylistic differences. More generally, the mean images highlight that digit 0 is slightly oval, digit 2 contains a small loop, digit 3 is vertically upright with a curved style on average, and digit 5 has a distinct flat top.

2.2 Mean & Standard Deviation of Zero Pixels

From Table 1, digit 1 has the highest mean of zero valued pixels (699.1), reflecting its simple structure, while digit 0 has the fewest (590.3), indicating that zeroes on average occupy the most image area. The remaining digits fall between 610 and 653 and, from fewest to most, follow the order: 8, 2, 3, 6, 5, 4, 9, 7.

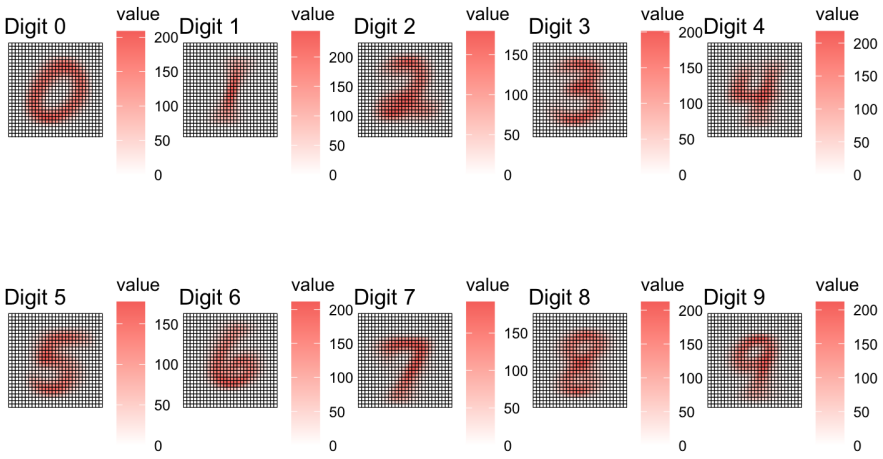


Figure 1: Mean Pixel Image Heatmap

For standard deviation, digit 1 shows the lowest variability (19.7), suggesting consistent length across writers. Digits 0, 2, 3, 5, and 6 have higher standard deviations (33–34), indicating greater variation in style, while digits 4, 7, 8, and 9 show moderate variability (26–31). Overall, the standard deviations are relatively small and differences across digits are modest.

Table 1: Zero Valued Pixels Summary Statistics

label	mean_zero_pixels	std_zero_pixels	label	mean_zero_pixels	std_zero_pixels
0	590.296	34.05154	5	632.390	33.56688
1	699.126	19.66580	6	625.644	33.21626
2	616.708	33.55346	7	652.934	26.86226
3	619.080	34.44012	8	610.398	31.60578
4	641.028	28.75944	9	641.712	28.62285

3 Methodology

This section summarizes the methods employed in the analysis: PCA, UMAP, and CCA.

3.1 Principal Component Analysis

PCA is a linear dimensionality reduction method that identifies the directions of maximal variance; known as principal directions. Equivalently, for data in R^p reduced to R^r, PCA selects the r dimensional subspace that minimizes reconstruction error, the difference between the original data and their projection onto the subspace spanned by the first r principal directions.

3.2 Uniform Manifold Approximation and Projection

UMAP is a nonlinear dimensionality reduction method that preserves the geometric structure of high-dimensional data by constructing a weighted k nearest neighbor graph and finding a lower dimensional embedding whose graph minimizes cross entropy with the original.

UMAP has two parameters: n_neighbors and min_dist. The n_neighbours parameter sets the size of the local region used to build the neighbor graph, with smaller values emphasizing local structure and larger values capturing broader patterns. The min_dist parameter controls how tightly points may cluster in the embedding: smaller values produce denser clusters, while larger values spread clusters apart. Notably, min_dist does not affect which points are considered neighbors; it only affects how the visualization of the.

3.3 Canonical Correlation Analysis

CCA identifies relationships between two datasets X and Y by finding linear combinations of their variables that are maximally correlated. Equivalently, it determines vectors u and v that maximize the correlation between the projections $u^T X$ and $v^T Y$.

4

How do the leading PCA directions differ across MNIST digit classes?

4.1 Motivation

To characterize the main characteristics of each digit class, we use PCA because it identifies orthogonal directions of variation and provides an interpretable linear decomposition, isolating the main modes of geometric variation within each class.

4.2 Approach

First, we applied PCA to each digit class' standardized pixel matrix and examined the first two principal components. Then, we generated PCA score plots (PC1 vs. PC2) to identify observations with extreme values along one component while remaining near zero on the other, selecting these images for visualization. For digits without such observations, we instead used two samples that are both large in the target PC but have positive and negative scores of similar magnitude on the other component. Next, we examined the positive and negative directions of PC1 and PC2, and finally, compared these directions with the selected observations to interpret the dominant modes of variation captured by each component.

4.3 Result

For all digits, the dominant modes of variation inferred from the eigendigits align with those observed in the extreme images. We therefore discuss the geometric interpretations of the positive and negative directions of PC1 and PC2 for each digit. Notably, the positive and negative directions represent opposite extremes of the same underlying pixel pattern: at each pixel location, one direction increases the pixel intensity while the other decreases it.

4.3.1 Zero (2(a) & 2(b))

PC1+ highlights pixels that make the zero more circular and adds thickness to the top-left and bottom-right regions, whereas PC1– produces a right leaning, more ovular zero. PC2+ yields an ovular upright zero, while PC2– produces a right leaning zero with added thickness in the bottom-left and top-right regions. These variations are complementary: PC1+ and PC2– thicken opposite diagonals. Additionally, PC1 eigendigits appear smoother and more symmetric, whereas PC2 eigendigits show less symmetry and smoothness.

4.3.2 One (2(c) & 2(d))

PC1+ corresponds to ones that tilt to the right, whereas PC1– captures upright ones. PC2+ reflects a stylized form with a small diagonal hook at the top and a horizontal band at the bottom, while PC2– corresponds to ones lacking these features.

4.3.3 Two (2(e) & 2(f))

PC1+ corresponds to twos with a narrow starting curve, no loop, and a thin, long upward curving tail, whereas PC1– emphasizes right leaning twos with a loop. Both PC2 directions represent loopless twos: PC2+ corresponds to shorter twos with a thicker tail, while PC2– captures larger twos with minimal curvature at the top.

4.3.4 Three (2(g) & 2(h))

PC1+ corresponds to a right-leaning curved three, whereas PC1– represents a left-leaning curved three. PC2+ highlights a stylized form with a flat top, while PC2– removes the pixels associated with this stylized shape.



((a)) Eigendigits 0 ((b)) Heatmap of 0 ((c)) Eigendigits 1 ((d)) Heatmap of 1
((e)) Eigendigits 2 ((f)) Heatmap of 2 ((g)) Eigendigits 3 ((h)) Heatmap of 3
((i)) Eigendigits 4 ((j)) Heatmap of 4 ((k)) Eigendigits 5 ((l)) Heatmap of 5
((m)) Eigendigits 6 ((n)) Heatmap of 6 ((o)) Eigendigits 7 ((p)) Heatmap of 7
((q)) Eigendigits 8 ((r)) Heatmap of 8 ((s)) Eigendigits 9 ((t)) Heatmap of 9

Figure 2: Eigendigits and Heatmaps of Digits 0–9 with Extreme PC Scores

4.3.5 Four (2(i) & 2(j))

PC1 places strong weight on the first vertical stroke and the lower half of the second: PC1+ produces a thicker, straighter form, whereas PC1– corresponds to a thinner, right-leaning version of the digit. PC2 emphasizes variation in the final vertical stroke: PC2+ yields a fully vertical form, while PC2– produces a more diagonal one.

4.3.6 Five (2(k) & 2(l))

PC1– corresponds to a stylized five with a straight horizontal bar, straight vertical line, and large loop, whereas PC1+ emphasizes an upward tilting horizontal bar and a smaller loop. PC2– captures another stylized form with an upward angled horizontal bar and a larger loop, while PC2+ reduces these stylized features.

4.3.7 Six (2(m) & 2(n))

PC1+ corresponds to sixes with a more vertical, thicker initial stroke and a larger, rounder loop, whereas PC1– reflects sixes with a thinner, more diagonal initial stroke and a smaller loop. PC2+ resembles the form seen in PC1– but has a slightly less diagonal initial stroke and a slightly larger loop, while PC2– removes the pixels defining this shape and adds intensity to the surrounding regions.

4.3.8 Seven (2(o) & 2(p))

PC1+ corresponds to sevens with a more diagonal vertical stroke and a straight horizontal bar, whereas PC1– reflects sevens with an upward-tilting horizontal bar, a stylized initial hook and a less angled vertical stroke. PC2 shows a similar pattern, however, PC2+ produces a straighter vertical line than PC1+, while PC2– yields a more strongly diagonal one.

4.3.9 Eight (2(q) & 2(r))

PC1+ corresponds to eights that tilt left, whereas PC1– represents eights that tilt right. PC2+ highlights a straight eight, while PC2– emphasizes an eight with a missing or broken middle section.

4.3.10 Nine (2(s) & 2(t))

PC1+ corresponds to nines with a smaller loop and a rightward tilt, whereas PC1– reflects upright, thicker nines with a large incomplete loop. PC2 produces forms similar to those in PC1, however, PC2+ yields a stronger right tilt, while PC2– captures a thinner nine and a smaller loop.

5

How do PCA and UMAP differ in their three dimensional representations of the global MNIST manifold, and how do UMAP's parameters affect digit separation?

5.1 Motivation

To understand the geometric relationships among handwritten digits, we compare the embeddings produced by PCA and UMAP. This comparison shows how methodological differences shape the perceived MNIST manifold and how each method organizes the digit classes. PCA provides a linear, global representation by preserving dominant variance, whereas UMAP yields a nonlinear, locally organized embedding based on neighborhood structure. Together, they offer complementary views of the dataset's global and local geometry.

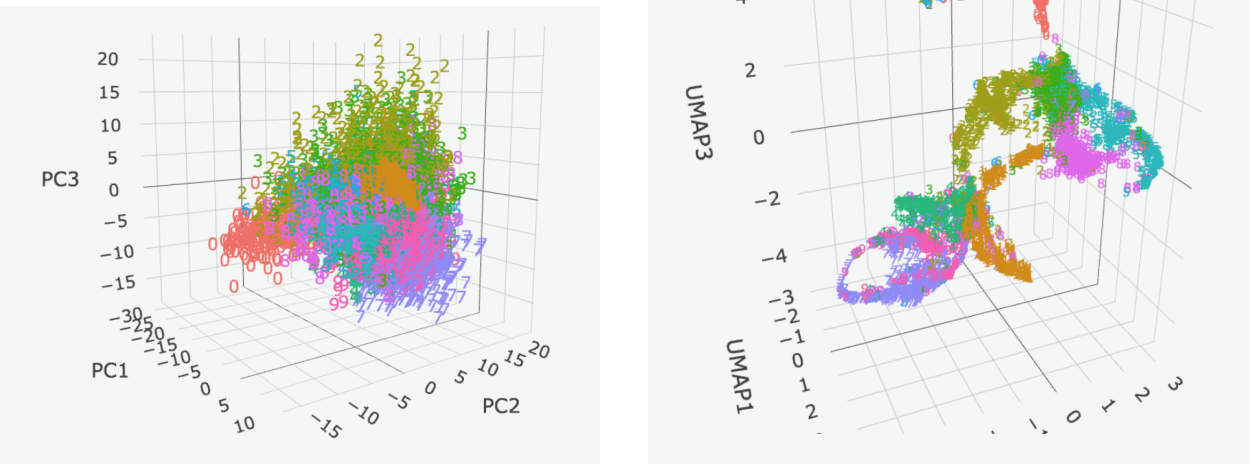
5.2 Approach

First, we applied PCA to the standardized MNIST dataset and projected all observations onto the first three principal components to visualize the dominant global variation. We then applied UMAP to obtain a three-dimensional embedding and systematically varied its parameters, n_neighbours and min_dist, to assess their effects on digit separation and cluster geometry. After identifying parameter settings that yielded the clearest separation, we compared the resulting UMAP embedding with the 3D PCA projection to evaluate how each method preserved digit clusters and inter-digit relationships.

5.3 Result

5.3.1 3D PCA on Global MNIST Dataset

Projecting the MNIST dataset onto the first three principal components (3(a)) results in a dense three dimensional embedding where many digit classes overlap. Some digits exhibit distinct patterns in how they vary along the principal components. Digit 0 spans negative PC1 with little variation along PC3, and digit 1 forms a compact cluster with minimal spread across components. Digit 2 reaches the largest positive PC3 values, whereas digit 7 occupies the most negative PC3 region. Digit 4 lies primarily in positive PC1 and negative PC3. Digits 3, 5, 6, 8, and 9 cluster near the center and exhibit substantial overlap. This overlap indicates that the first three principal components do not capture the variation needed to distinguish these classes. This suggests that the relevant discriminative structure may lie in higher order components or may reflect nonlinear relationships that cannot be recovered through a low dimensional linear projection.



((a)) 3D PCA on Global MNIST Dataset

((b)) Optimal UMAP

Figure 3: Comparison of PCA and UMAP Embeddings

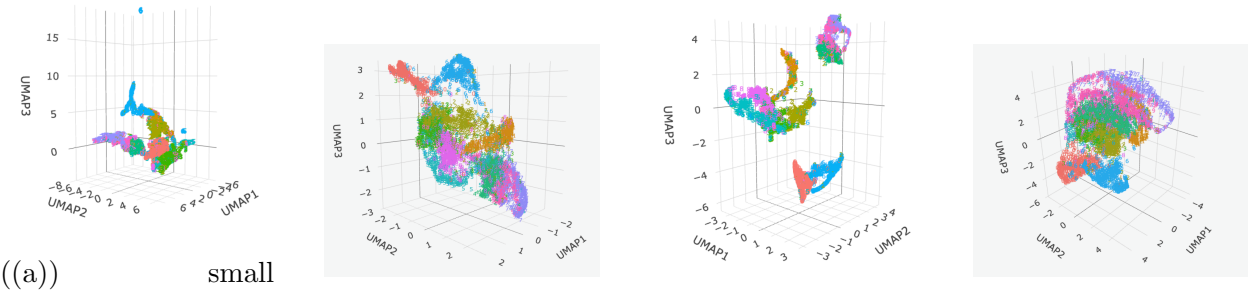
5.3.2 UMAP Parameter Exploration

When n_neighbours is too small (3(a)), UMAP focuses on local structure, causing each digit to fragment into small subclusters rather than forming coherent class level groups. When n_neighbours is too large (3(b)), the embedding becomes overly global and digit classes begin to overlap. For min_dist, small values (3(c)) pull same digit points tightly together, producing compact clusters that obscure within class variation, whereas large values (3(d)) spread each digit more broadly, increasing visible variation but also bringing neighboring digits closer together and causing overlap.

5.3.3 3D UMAP on Global MNIST Dataset

From experimenting with different parameter settings, we find that the configuration that produced the strongest digit separation is n_neighbors = 21 and min_dist = 0.09.

Under these settings, UMAP (3(b)) forms distinct clusters for all digits and separates the MNIST classes effectively in three dimensions. Each digit occupies its own identifiable region



((a)) small n_neighbor ((b)) large n_neighbor ((c)) small min_dist ((d)) large min_dist

Figure 4: Effect of UMAP Parameters: Varying n_neighbors (5, 200; min_dist = 0.1), Varying min_dist (0.01, 0.5; n_neighbor = 20).

in the embedding. However, there are steveral cases of misplacement that occur, and many are reciprocal: when examples of digit A appear inside digit B's cluster, instances of digit B will also appear within digit A's region. Notable examples include cross placements between 0 and 6, as well as interactions among 4, 7, and 9. Additionally, one end of the clusters for digits 3, 5, and 8 overlaps, indicating shared structural variation that UMAP cannot fully disentangle in three dimensions.

5.3.4 PCA vs. UMAP: Digit Separation in Three Dimensions

In conclusion, in three dimensions, UMAP provides substantially better digit separation than PCA. PCA produces broad, overlapping clusters, particularly among digits 3, 5, 6, 8, and 9, indicating that the true digit manifold is not three dimensional or not well approximated by a linear subspace. UMAP, by contrast, forms distinct clusters for all digits and achieves much clearer separation, although cross placement still occurs for geometrically similar digits. Additionally, digits 3, 5, and 8 exhibit partial cluster overlap, reinforcing that the underlying manifold likely exceeds three dimensions. Notably, UMAP separates digits 6 and 9, which PCA could not, since the differences between these digits arise from nonlinear shape variations rather than linear trends.

6

How do canonical correlations between the top and bottom halves of digits differ, and what do these differences reveal about the horizontal symmetry of each digit?

6.1 Motivation

It is obvious for the human eye to observe that some digits (e.g. 0) are horizontally symmetric while other digits (e.g. 2) are not. Instead of relying on visual intuition, we want to turn this into a numerical scale. CCA is an ideal tool for this setting because it measures the maximum linear association between two high-dimensional sets. Thus, we use CCA to examine and quantify how the horizontal symmetry differs for the top and bottom halves of each digit.

6.2 Approach

We use CCA to investigate the linear relationship between the upper and lower halves of the same digit class. We choose digits 0, 2, 7 to investigate. For each digit class, we split it horizontally: columns 1-392 form the upper half and columns 393-784 form the lower half. Thus we have two data blocks for each given digit: X refers to the top half and Y refers to the bottom half. Before applying CCA, we first apply PCA separately to the upper and lower halves for each digit and retain the 50 leading principal components, since CCA cannot perform on a high-dimensional dataset. These PCs approximately explain 90% of the variance in each half. We apply CCA on the 50-dimensional PC scores matrices X_r and Y_r of each digit class. For each digit, we have ordered canonical correlations $\rho_1 \geq ... \geq \rho_{50}$. We then sum the first 50 canonical correlations of each given digit: $S = \sum_{k=1}^{50} \rho_k$. A higher cumulative value indicates that the two halves have a stronger liner relationship, while a lower cumulative value indicates weaker correspondence, which confirms the visual asymmetry.

6.3 Result

The cumulative values for digits 0, 2, 7 are 22.34629, 18.31396, and 19.6098. Digit 0 has the highest cumulative correlation, as its shape is almost perfectly symmetric. Digit 7's cumulative value ranks in the middle. Although its overall shape is asymmetric, both halves share a dominant vertical line. Digit 2 has the lowest cumulative value because its top half is a curved loop and its bottom half is nearly horizontal. The results help to quantify what our eyes suggest about the horizontal symmetry.

7 Discussion

We will discuss the limitations and future directions of our project.

7.1 Limitations

For Research Question 1, PCA is limited by its linearity and cannot capture nonlinear variation within digit classes. Its interpretation also relies on visually selected extreme score images, introducing subjectivity. For Research Question 2, the choice of optimal UMAP parameters was based on visual inspection rather than a quantitative criterion, which may affect the accuracy of the comparison with PCA. In addition, UMAP embeddings vary across runs due to random initialization, and we did not account for this variability. For Research Question 3, CCA captures only linear dependence between the top and bottom halves of digits and may underestimate nonlinear symmetry. In addition, we examined symmetry for only a subset of digits.

7.2 Future Work

For Research Question 1, future work could include applying nonlinear dimensionality reduction method such as Kernel PCA and probabilistic approaches such as Factor Analysis to uncover within digit variation that linear PCA cannot capture. For Research Question 2, a quantitative measure of digit separation can be implemented such as evaluating class separation with a k nearest neighbours classifier on the 3D embeddings. For Research Question 3, the symmetry analysis can be broadened to all digit classes and expanded to include diagonal and vertical forms of symmetry, providing a more comprehensive understanding of digit structure.

8 Reference

The authors use AI tools to help with writing, phrasing, and organizing their report. Parts of the code in this report were adapted from the course material practice5-pca.html, practice6-pca-variants.html, and practice7-cca.html.